

CWC 2.0 ESM — Frontend Map

Root: frontend/ · **Stack:** React 19 + TypeScript + Vite + React Router. All ESM pages eagerly imported, wrapped in ProtectedRoute.

1. Routing (frontend/App.tsx, lines 249–337)

ESM core routes (CRM and credit excluded):

Path	Component	Guard
`/`	`Dashboard`	ProtectedRoute
`/hr`	`HRServices`	ProtectedRoute
`/it`	`ITSupport`	ProtectedRoute
`/finance`	`GroupFinance`	ProtectedRoute
`/esm`	`ExecutiveServices`	ProtectedRoute
`/my-requests`	`MyRequests`	ProtectedRoute
`/request/:id`	`RequestDetail` (+ `ErrorBoundary`)	ProtectedRoute
`/it/hardware`	`Navigate → /it`	—
`/agent`	`AgentDashboard`	ProtectedRoute
`/reports`	`Reports`	`report:read`
`/insights`	`Insights`	`report:read`
`/search`	`SearchResults`	ProtectedRoute
`/kb`, `/kb/:slug`	`KnowledgeBase`, `ArticleDetail`	`isFeatureEnabled('kb')` + `kb:manage`, else redirect `/`
`/approvals`	`ApprovalCenter`	ProtectedRoute
`/inbox`	`UnifiedInbox`	ProtectedRoute
`/announcements`, `/:id`	`Announcements`, `AnnouncementDetail`	ProtectedRoute
`/admin/announcements`	`AnnouncementsManage`	`announcement:write`
`/assets`	`AssetManagement`	`asset:read`
`/admin/settings`	`AdminSettings`	`admin:access`
`/admin/workflows`	`WorkflowList`	`admin:access`
`/admin/workflows/:workflowTypeId/versions/:versionId`	`WorkflowDesigner`	`admin:access`
`/admin/audit`	`AuditTrail`	`admin:access`
`/:deskType/:deskId/create/:categoryId`	`CreateRequest`	ProtectedRoute
`/*`	`NotFound`	—

Guards (src/components/ProtectedRoute.tsx): checks isAuthenticated (→ /login), optional requirePermission (OR), requireAllPermissions (AND), requireDepartment, legacy requireAdmin.

2. Pages (frontend/pages/)

- `Dashboard.tsx` — Personalized home: greeting, per-desk category tiles (IT/HR/Finance), recent announcements, my recent requests. Uses `serviceDeskService`, `requestService`, `announcementService`.
- `HRServices.tsx` / `ITSupport.tsx` / `GroupFinance.tsx` / `ExecutiveServices.tsx` — Service-desk landing pages; fetch the desk by code (HR/IT/FINANCE/ESM) via `serviceDeskService.getAllServiceDesks()`, render categories as cards → `/hr|it|finance|esm/:deskId/create/:categoryId`.
- `MyRequests.tsx` — User's request list with status/desk filters, pagination → `/request/:id`.
- `CreateRequest.tsx` — 3-step wizard (see §3).
- `RequestDetail.tsx` — 2-pane request cockpit (see §3), uses `useRequestDetail` hook.
- `AgentDashboard.tsx` — Support queue for ADMIN/AGENT: assigned/open/unassigned queues, SLA stats, filters.
- `ApprovalCenter.tsx` — Tabbed approval hub (`all/itsm/credit`); ESM half uses `approvalService`, credit half uses `creditService`.
- `ApprovalQueue.tsx` — Pending-approval queue: search, desk/priority filters, bulk approve/reject, pagination.
- `UnifiedInbox.tsx` — Consolidated notification inbox (`notificationService`).
- `Reports.tsx` — Operational reports (`reportsService`: `summary/status/desk/priority/agent-workload/SLA-status`).
- `Insights.tsx` — Trend/analytics (`insights.service`).
- `KnowledgeBase.tsx` / `ArticleDetail.tsx` — Article list + single article (mark helpful/unhelpful).
- `SearchResults.tsx` — Global search (`search.service`).
- `Announcements.tsx` / `AnnouncementDetail.tsx` / `AnnouncementsManage.tsx` — Browse/view/admin CRUD announcements (`RichTextEditor`).
- `AssetManagement.tsx` — Large (1590-line) asset admin: list/filter/search, assign/return, user assignments, CSV/XLSX import (parse → validate → commit) + CSV export.
- `WorkflowList.tsx` — Versioned workflow list (search + create-draft dialog).
- `WorkflowDesigner.tsx` — Visual graph editor (see §6).
- `AdminSettings.tsx` — Tabbed admin console (see §4).
- `AuditTrail.tsx` — Audit log browsing.

3. Request lifecycle UI

CreateRequest wizard (`src/components/create-request/`):

- `useCreateRequestWizard.ts` — core state hook. Steps type → details → review. Loads categories + request types via `serviceDeskService`; entity options via `entityService.listActiveEntities()`; CEO/Group DCEO via `apiclient.get('/users/executives?role=CEO,GROUP_DCEO')`. Parses the request type's **form schema** (`parseFormConfig(type.formConfig)`), auto-generates summaries per request-type code, computes auto-confidentiality, blocks role-gated types.
- `StepRequestType.tsx`, `StepDetails.tsx` (dynamic form from schema), `StepReview.tsx`, `WizardStepper.tsx`, `RecentServices.tsx`, `useDraftSave.tsx` (`localStorage` draft autosave).
- Submit calls `requestService.createRequest({...})` after `validateFormValues`; navigates to `/request/{referenceNumber}`.

FormBuilder (src/components/FormBuilder.tsx) — Admin builder for request-type custom-field schemas: drag-to-reorder (dnd-kit), field types text|textarea|select|date|number|currency|file|entity|ceo-select, per-field required, **conditional visibility** (showWhen rules). Feeds onSave(fields) → persists formConfig.

RequestDetail (pages/RequestDetail.tsx + src/components/request/, src/components/request-detail/):

- useRequestDetail.ts hook (750 lines) — all state/handlers: fetch, updateStatusDirectly, handleStatusChange, resolution/rejection, interviews, HR screening, LOA approve/issue/accept, onboarding/offboarding phase advance + completion, candidate/resume, route-to-CEO/manager/GroupDCEO.
- **Left pane:** RequestFormFields, HiringWorkflowPanel, OnboardingDashboard/OffboardingDashboard, ActivityFeed, RequestHeader.
- **Right pane:** WorkflowCockpit — WorkflowStepper, SLAIndicator, ApprovalChain, ApproverPicker, actions/transitions, attachments, assign agent, PDF export.
- **Transition modals** (src/components/request-detail/): GenericTransitionModal, DecisionPanel, WorkflowActionModal, AssignAgentModal, AssignToDropdown, ParticipantsSection, CustomFieldsPanel, BatchUploadModal, plus desk-specific (FinDecisionModal, CfoDecisionModal, CeoDecisionModal, RouteToCeoFinModal, HardwareOrderedModal, HardwareReceivedModal, SoftwareProvisionedModal, MarkJobPostedModal, ProcurementModal, PendingInvoiceModal, PaymentDoneModal, CloseTicketFinModal, CompleteDeliveryModal, UploadSignedLOAModal, UploadLOAModal, UpdateScreeningModal, UploadResumeModal).
- src/components/request/modals/ — ResolutionModal, RejectionModal, CompleteOnboardingModal, ScheduleInterviewModal, EditInterviewModal, InterviewFeedbackModal, LOAApprovalModal, CEODecisionModal, ManagerDecisionModal.
- Attachments via requestService; PDF export via exportPdf + pollPdfJob. SLA shown by SLAIndicator (uses slaDueAt, slaPausedAt, slaPauseDurationMs).

4. Components (src/components/)

- **Layout** (layout/): LeftRail.tsx (grouped sidebar), TopBar.tsx, MobileDrawer.tsx. navConfig.ts — buildNavLinks(user) returns groups primary | service-desks | tools | admin, permission/role/feature-flag gated (Approvals if request:approve/credit:approve; Agent for ADMIN/AGENT; KB if kb flag + kb:manage).
- **Guards/context:** ProtectedRoute.tsx, ErrorBoundary (withErrorBoundary.tsx).
- **Request dashboards:** OnboardingDashboard.tsx, OffboardingDashboard.tsx (phase/task trackers).
- **Approvals:** EntityApprovalsPanel.tsx (entity-scoped approvals; standalone — not imported by a live page).
- **FormBuilder:** FormBuilder.tsx + admin/FormBuilderModal.tsx.
- **Notifications:** NotificationDropdown.tsx (driven by NotificationContext).
- **Admin settings:** admin/adminConstants.ts — ADMIN_TABS grouped Configuration | Workflows | Appearance: Service Desks, User Accounts, Permissions, Entities, Email Notifications, Onboarding Tasks, Offboarding Tasks, Runtime Transitions, Request Statuses, SLA Escalation, ESM Settings, Scheduler, Banner & Branding.
- **Workflow designer:** workflow/ — WorkflowCanvas, StatusNode, StatusPalette, NodeInspector, EdgeInspector, ValidationPanel, PublishDialog, VersionHistoryPanel, CreateDraftDialog, WorkflowListCard.

- **UI kit** (ui/): Button, Card, Modal, Drawer, Tabs, Tooltip, Skeleton, EmptyState, StateBadge, RichTextEditor, Combobox, AutosaveTextField, EnvironmentBanner, OutOfOfficeModal, PolicyExplainer.

5. Services & types (src/services/)

- request.service.ts (requestService) — getAllRequests, getRequestById, getAvailableTransitions, createRequest, updateRequest, deleteRequest, getRequestActivities, addActivity, attachment upload/download/delete, assignRequest, updateStatus, participants, getRecentServices, exportPdf, exportXlsx. Types: AvailableTransition, RequestParticipant, CreateRequestData, RequestFilters.
- serviceDesk.service.ts — read (desks/categories/request-types), admin CRUD, deactivation-impact, agents, **escalation/SLA rules** (getEscalationRules, CRUD).
- approval.service.ts (approvalService) — CEO/GroupDCEO/manager decisions, markJobPosted, resumes/candidates, getPendingApprovals, bulkAction, getPolicyExplanation.
- workflow.service.ts (workflowService) — legacy linear WorkflowType/Step CRUD.
- workflow-version.service.ts (workflowVersionService) — **versioned graph API**: listWorkflows, listVersions, createDraft, getVersion, updateNodes/updateEdges (batch PATCH), replaceGraph, validateVersion, publishVersion(statusRemap), rollbackVersion, discardDraft. Types: WorkflowGraph, GraphNode, GraphEdge, ValidationResult/ValidationFinding, RemapPlan/RemapEntry, PublishResult.
- asset.service.ts — listAssets, getAsset, createAsset, updateAsset, deleteAsset, assignAsset, returnAsset, listActiveAssignments, getAssetsByUser, importAssets (+ parse/commit), exportAssetsCsv.
- kb.service.ts — getArticles, getArticleBySlug, markHelpful.
- notification.service.ts — getNotifications, getUnreadCount, replayAfter, markAsRead, markAllAsRead, deleteNotification.
- reports.service.ts — getSummary, getByStatus, getByServiceDesk, getByPriority, getAgentWorkload, getSlaStatus.
- systemSetting.service.ts — getEsmDceoThreshold / setEsmDceoThreshold (Group DCEO travel threshold, default 50000).
- admin.service.ts — users, roles/permissions, listWorkflowTransitions CRUD, notification templates, email toggle, onboarding IT agent.
- **Other**: announcement.service, search.service, insights.service, auditLog.service, interview.service, screening.service, loa.service, finance-workflow.service, it-workflow.service, esm-workflow.service, requestStatusService.

6. Workflow Designer UI

Graph-based editor (pages/WorkflowDesigner.tsx + src/components/workflow/ + src/hooks/useWorkflowGraph.ts).

- Route: /admin/workflows/:workflowTypeId/versions/:versionId (guard admin:access). Read-only when version ≠ DRAFT.
- **Canvas** (WorkflowCanvas.tsx): built on @xyflow/react (React Flow). Nodes are StatusNode; edges are workflow type.

- **Authoring:** StatusPalette adds nodes from the governed status catalog; NodeInspector edits statusCode/label/icon/isInitial/isFinal/slaPause/displayOrder; EdgeInspector edits transition label/requiresComment/autoAssignRole/autoAssignUserId/allowedRoles/allowedExecutiveRoles.
- **State & saving** (useWorkflowGraph.ts): React Flow state + dirty flag + **500 ms debounced autosave** via workflowVersionService.replaceGraph(versionId, graph), then re-runs validateVersion. Converts between React Flow and domain workflowGraph via utils/workflowLayout.ts.
- **Validation:** ValidationPanel surfaces blocking and warnings from validateVersion; blocking (except STATUS_IN_USE_REMOVED) disable Publish.
- **Publish** (PublishDialog.tsx): if removed statuses still hold live requests, step 1 collects a **status remap** from RemapPlan; step 2 confirms warnings + request count, then publishVersion(versionId, statusRemap).
- **Versioning:** VersionHistoryPanel lists DRAFT/ACTIVE/ARCHIVED; supports open-version, **rollback** (from ARCHIVED), **discard draft**. WorkflowList.tsx lists workflows with badges + CreateDraftDialog.

7. Notes

- **i18n:** ESM pages are hard-coded English (only pages/Dashboard.tsx uses useTranslation from the global i18n config).
- **Sentry:** only at app level via global ErrorBoundary; no ESM page calls Sentry directly.
- ApprovalCenter / MyApprovals / Dashboard **mix in credit/CRM UI**; this doc covers their ESM-relevant parts.